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RESEARCH ARTICLE

Grid-Scale Battery Systems for VAR Support in Con Edison Distribution Substations via Bi-Directional Smart Inverters

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ABSTRACT Battery Energy Storage Systems (BESSs) play a pivotal role in enhancing the grid's reliability by integrating Distributed Energy Resources (DERs) and offering a range of services, notably including reactive power (VAR) support through the use of bidirectional smart inverters. According to the literature review carried out and presented in this paper, it is evident that reactive power control for DER applications and grid have been examined in various studies. However, there is still a need to perform a thorough and explicit investigation on the VAR support impact on battery lifetime— we attempt to fill this gap with this study. This work aims to explore the effects of reactive power compensation provided by BESS smart inverters on the system's lifespan, and to identify strategies for mitigating any potential negative impacts. For this purpose, a simulation was conducted on a 7.5MW/30MWh BESS connected to the grid at a Con Edison substation. The analysis revealed the significance of filter design on the DC side in mitigating battery current flow, thereby influencing the State of Charge (SoC) and extending the battery's life. Implementing adequately sized inductive filters on both AC and DC sides proved effective in controlling battery current fluctuations while keeping the total harmonic distortion (THD) within acceptable ranges. Various scenarios were examined to ascertain the optimal inductive filter size that balances the BESS's real and reactive power outputs. Additionally, simulations incorporating real-world grid elements at the substation were performed, including an evaluation of the combined effects of BESS VAR support with substation capacitor banks and shunt reactors. The findings suggest that inductive filters sized at 1mH and 5mH for both AC and DC interfaces can facilitate VAR support with minimal detrimental effects on BESS longevity. Moreover, a load flow analysis, integrating the BESS with Con Edison's existing substation infrastructure, demonstrated the potential for bus voltage regulation and improvement by adjusting the BESS inverter's reactive power settings.

INDEX TERMS Battery energy storage system (BESS), Bi-directional smart inverter, filter design for BESS, lifetime of BESS, load flow analysis, reactive power (VARs), total harmonic distortion (THD).

NOMENCLATURE

BESS	Battery Energy Storage System.
CAL	Calendar Lifetime.
CYC	Cycle Life Time.

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DERs	Distributed Energy Resources.
DQZ	Direct Quadrature Zero.
D-STATCOM	Distribution Static Synchronous Compensator.
IGBT	Insulated Gate Bipolar Transistors.
LFP	Lithium Iron Phosphate.
MCS	Mechanically Switched Capacitors.
MG	Micro Grid'.
NYISO	New York Independent System Operator.
PCU	Power Converter Unit.
PLL	Phase Locked Loop.
PWM	Pulse Width Modulation.
P	Active Power.
P-f	Power frequency.
$PD_{\beta,k}^C$	Charging of active power (Watt) of battery β at time 'k'.
PB^C	Active charging power of battery.
PB^D	Active discharging power of battery.
$PB_{\beta,k}^D$	discharging of active power (Watt) of battery β at time 'k'.
Q	Reactive Power.
QB^C	Reactive charging power of battery.
QB^D	Reactive discharging power of battery.
$QB_{\beta,k}^C$	Charging of Reactive power of battery β at time 'k' (VAR).
$QB_{\beta,k}^D$	Discharging of reactive power (VAR) of battery β at time 'k'.
Q_{c1}	Fundamental Component of reactive power.
Q-V	VAR-Voltage.
r_{AC}	AC side resistance.
r_{DC}	DC side resistance.
SoC	State of Charge.
SOHc	Cycle-based State of Health.
SOHt	Calendrical aging.
STATCOM	Static Synchronous Compensator.
SVC	Static VAR Compensator.
THD	Total Harmonic Distortion.
VAR	Reactive power.
VSI	Voltage Source Inverter.

SYMBOLS

B	Set of batteries within system.
d-q	direct axis-quadrature axis.
f_s	Switching frequency.
i_a, i_b, i_c	Current in phase a,b and c respectively.
I_{c1}	Fundamental Component of inverter side Current.
I_d	Direct axis current.
I_q	Quadratic axis current.
ΔI_L	Current ripple in inductor.
SB_{β}^{rated}	Apparent Power rating (VA) of battery β .
u_d, u_q	Control inputs.

$[v]_{abc}$	Inverter output voltage.
V_{c1}	Fundamental component of voltage at inverter side.
V_d	Direct axis voltage.
V_{IN}	Voltage at input of DC side Inductor.
V_{OUT}	Voltage at output of DC side Inductor.
V_q	Quadratic axis voltage.
$[v_s]_{abc}$	Three phase grid voltage.
ω	Angular frequency.
I_{SOH}	Index to calculate impact of VAR support on health of battery.
K	Set of time periods.
K_i	Integral Controller gain.
K_p	Proportional Controller gain.
L_{AC}	AC side Inductance.
L_{DC}	DC side Inductance.
I_{OUT}	Current in DC side inductor.

I. INTRODUCTION

Battery Energy Storage Systems (BESSs) are emerging as transformative forces within the power sector, recognized for their critical role in facilitating the widespread adoption of high integration of renewable energy [1]. The unique intermittent nature and instability of renewable energy introduce significant stability challenges to power systems, creating mismatches between energy production and consumer demand. Thus, energy storage technologies serve as a pivotal tool for ensuring the reliability and efficiency of renewable energy [2]. Additionally, advancements in smart control technologies have empowered Distributed Energy Resources (DERs) to offer more adaptable load-balancing solutions, effectively managing the fluctuations in renewable energy output [3], [4]. By deploying BESSs in proximity to consumer locations, they also enhance the resilience of smart cities against major operational disruptions. It is widely acknowledged that BESSs will be essential in the development of future power grids [5]. Despite the gradual reduction in the cost of BESS, it is still relatively high. [6].

Identifying strategies to leverage the multifaceted advantages of Battery Energy Storage Systems (BESS) installations has become increasingly important [7]. These benefits include frequency regulation [8], flexible ramping [9], load following, deferring upgrades for transmission and distribution systems [10], [11], capacity firming, reducing curtailment, providing backup power for emergencies or enabling selective islanding [12], ensuring a reliable power supply to isolated grids, peak shaving [13], and offering voltage/reactive power support [14], [15], [16]. This study focuses on examining a specific real-world BESS deployment aimed at delivering reactive power support, alongside its principal functions of peak shaving and participation in the NYISO market. Through the use of smart inverters, BESSs are capable of supplying reactive power for grid voltage support, comparable to shunt reactive devices such as Mechanically Switched

Capacitors (MSC), Static VAR Compensators (SVC), and D-STATCOM. Like D-STATCOM systems, BESS inverters have the capability to dynamically regulate voltage.

In the United States, utility companies such as Con Edison are setting aggressive targets for the deployment of Distributed Energy Resources (DERs) and Battery Energy Storage Systems (BESSs). There's a significant opportunity to utilize BESS inverters for VAR support in addition to their core functionalities, which could mitigate the need for additional reactive power compensation devices like D-STATCOM and MSC, thereby reducing costs. However, it's essential to conduct preliminary studies to evaluate the impact of VAR support on the longevity of BESS. While there's a wealth of research on the use of DER-connected inverters and devices like DSTATCOM for VAR compensation [17], [18], further investigation into the application of grid-connected BESS for VAR support and its effects on battery life is necessary.

This gap in research highlights the need for a detailed study on battery degradation and the implications of using large-scale BESS for VAR support. The methodology and system under analysis are detailed in Section II. Section III presents and discusses the findings of this research. The outcomes of the steady-state load flow analysis are disclosed in Section IV. Lastly, Section V offers a summary of the pivotal conclusions drawn from this study.

A. LITERATURE REVIEW

The phase difference between terminal voltage and output current can be controlled by inverters through Pulse width modulation (PM) and Insulated gate bipolar transistors (IGBTs) switching. Hence, the operation at lagging or leading power factors can be achieved while complying with the apparent power capacity.

They can operate at any leading or lagging power factor if their apparent power capacity is obeyed. BESS systems, utilizing their inverters, provide a flexible source of reactive power, as seen in Fig. 1. Inverter technology for BESS and solar systems are similar. Realizing this, ongoing research suggests that the efficiency of inverters is the main factor for the assessment of compensation of reactive power from PV inverters. However [19], reactive power support provided by the battery system alone is not discussed. It was determined that various control schemes can be implemented in solar PV inverters to get reactive power support. However, the capability to provide reactive power support is constrained if compared to synchronous generators. Thus, a battery storage system can be used at the DC link of the solar PV inverter. Increased reactive and active power support can be obtained by linking BESS with the DC side of the solar PV inverter [19].

Additionally, designated power plants utilize capacitor banks, shunt reactors, and onload tap changers to produce reactive power to compensate the grid reactance. A general comparison of the various technologies conventionally

used for VAR support in addition to generators, capacitor banks, and onload tap changers, has been presented [20]. Table 1 presents the comparison. These technologies include Synchronous Condensers (SC), Static VAR Compensators (SVC), and STATCOMs, as shown in Table 1. Strategic deployment of energy storage at central locations or at the grid can displace these devices. Various storage systems can also be aggregated near large loads using the distributed technique [21]. There are issues related to discharging/overcharging of batteries due to an unbalanced State of Charge (SOC). Violations in the voltage limits are observed due to the disproportional sharing of reactive power. The above-mentioned issues can be resolved by using a decentralized control technique in islanded MG comprising of BESS and intermittent DERs [22].

Therefore, power-frequency (P-f) droop-based decentralized techniques for SoC balancing and reactive power-voltage (Q-V) droop-based decentralized strategies for sharing reactive power were established in [22]. The objective was the regulation of the SoC of battery (hence, real power output), and proportional dispatch of reactive power among battery systems, respectively. The authors of [22] have not considered the life cycle of the battery but suggest considering it for future research. According to research findings in [23], injection of reactive power by battery and PV inverters increases power losses. Pricing of reactive power is computed by incorporating this incremental cost occurring due to losses [23]. Moreover, the assessment of the impact of reactive power support on the lifetime of the inverter was also carried out. This impact on the lifetime was considered while evaluating the cost of reactive power from the PV inverters. Finally, it was concluded from the results that the consideration of the reduced lifetime of inverters did not offset the economic benefits of reactive power support from PV inverters [23].

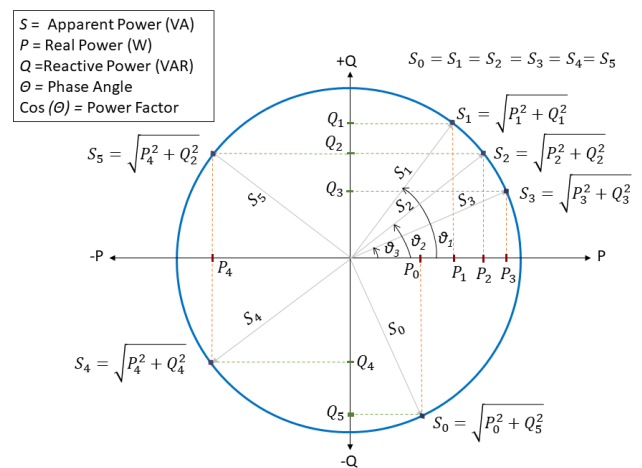


FIGURE 1. Real and rated apparent power of inverter.

The reactive and active power control in view of the micro-grid was examined in [24]. Voltage and active power control by battery system were suggested to minimize the adverse

TABLE 1. Comparison of technologies for VAR support.

Technology	Cons	Pro ^a
SC	• High losses & noise • Slow response	Provides continuous reactive power
STATCOM	• High investment cost • Complex control	• More accurate Regulation • Higher operational flexibility
SVC	Generates harmonics	• Thyristor control

effect of microgrid interconnection and for mitigating the voltage fluctuations in the distribution feeder. It was evident from the experimental results that microgrid voltage can be regulated by deploying reactive and real power control of the battery system. Frequency and voltage droop control of battery system showed that a smaller value of droop coefficient could perform smooth regulation of the microgrid voltage [24], [25].

It was studied in [26] that low voltage issues at secondary feeders can be resolved by using battery system for load balancing and reactive power support. The costs were not increased. In their model, the rating of converter of battery system was assumed to be 37% of capacity of substation transformer. The results showed that BESS can provide an extra range for reactive power compensation and current and voltage balancing. There was no effect on the service life of storage and BESS was not oversized according to the findings of [26]. Control algorithms for equipment can be developed and BESS can be designed using these results.

Cost-benefit analysis of BESS has been presented in [27]. The objective of the analysis was to study lifetime of battery considering SOC within the operating range of 15-85%. Subsequently, calendrical aging (SOH_c) and state of health based on cycle (SOH_{cy}) were estimated. In [28], Cycle lifetime (CYC) and Calendar lifetime (CAL) were examined to determine the lifespan of the lithium battery system [28]. Various methodologies based on deep learning methods and ultrasonic reflection waves have also been proposed for accurate estimation of SOC of batteries [29], [30].

In [31], an analysis has been presented to investigate the impact on the degradation of the battery due to its utilization for frequency response. SOC profile is given as the input to the model and temperature is kept constant at 25°C. The authors in [32], created a model known as the economic-degradation model for quantification of the impacts of various operational policies (primarily focused on limiting SOC to recommended levels) on the lifespan of energy storage systems and degradation, multi-service portfolios, and gross revenue. It has been demonstrated in [32] that battery lifetime can be reduced if the utilization of storage systems is increased. The approaches have been presented to examine the effects of frequency control, peak shaving, and energy arbitrage services on battery degradation [33], [34], [35].

It is evident from the literature review that previous studies in the literature have not focused on determining the effect of

reactive power support provided by the battery systems on the degradation of batteries.

B. CONTRIBUTION

According to the review presented here, reactive power control for DER applications and grid have been examined in various studies. However, there is still a need to carry out a thorough and explicit investigation on the VAR support impact on battery lifetime— we attempt to fill this gap with this study.

This paper has investigated the expansion of a utility-owned BESS system (located at an Area Substation) use cases beyond active power contribution. This BESS is located in a dense urban area in NYC where space constraints limit the size and amount of BESS that could be deployed. Therefore, it is imperative that when deployment of a BESS becomes feasible, we maximize and stack all the benefits. Furthermore, from the utility perspective, the use of battery for VARs support is becoming just as important as peak shaving for the following reasons:

- a. Maintaining the power quality of the distribution system
- b. Improvement in distribution system voltage during limit preemptive 5-8% voltage reduction at the substation (operational measure)

However, there are many unknowns associated with the advanced use of a 4-quadrant smart inverter for voltage/VAR injection with protection capability. With this in mind, the contributions of this study are as follows:

1. We demonstrated that with the proper inductive filters in place, VARs compensation has no implications on the active power (charge or discharge) of the battery. Therefore, these commands can occur in parallel and will be administered by the operators through the BMS (battery management system).
2. It has been determined that there are no thresholds for active or reactive power beyond the inverter size.
3. By limiting the direct current flowing out or into the battery during VAR operations, we demonstrated that there would minimal increase in degradation (cycling) and as such no impact on the BESS lifetime. This is of utmost importance for the utility as they will not have to renegotiate performance contracts for this system or pay for additional system augmentation.
4. It can be argued that reactive power support is already being achieved in the grid using capacitor banks, STATCOMS, and shunt reactors. So why use the BESS for this service? The key thing to note is that these are binary systems, that is, either on or off at the full rated capacity. Whereas the BESS allows operators to tailor the VARs response to evolving grid conditions without installing or maintaining additional equipment. Hence, with a single BESS installation, deployment of STATCOM can be avoided and the same functions can be achieved year-round vs. certain times of the

year. furthermore, the BESS will not have to be resized or warranty structure will not have to be renegotiated beyond the active power requirements.

II. METHODOLOGY

A. SYSTEM DESCRIPTION

A Lithium Iron Phosphate (LFP) battery system having the capacity of 7.5MW/30MWh has been considered for this study. This system is installed at Con Edison (utility) substation. A new distribution feeder of 33 kV will be used to connect the BESS with the substation cubicle.

There are 11 Tesla mega-packs each with a capacity of approximately 3.12 MWh. There are 19 battery modules in each megapack, which are connected in parallel, forming a string. A DC/AC converter is connected to each string (See Fig. 2). Subsequently, each mega pack is connected to a DC/AC converter. Con Edison provided the parameters for modeling the batteries and inverters. Three 33 kV/480V distribution transformers with a capacity of 3.3 MVA each, are connected to a 480 V bus. The 11-mega packs are also connected with this common bus. The concept of open-air design is mimicked in the layout of existing substation equipment as depicted in Fig. 2.

B. APPROACH/METHODOLOGY

A simulation model was created to investigate the BESS and its integration with the substation, as well as to test VAR support capability. Specifically, existing information and data were gathered, assembled, and evaluated. A transient simulation model was then developed in the MATLAB/Simulink software environment.

For the system integration and controls, the model has already been validated in our published papers [36], [37]. The simulation model replicated a typical single-line diagram of the Con Edison substation, battery specifications, and short circuit impedance. The developed model was utilized to investigate the reactive power support capabilities of BESS and the possibility of using existing shunt reactors and/or capacitor banks at the substation to provide BESS VAR support on the associated distribution feeder. Table 2 depicts the parameters used for modeling the system. Four different scenarios have been simulated and the methodology corresponding to each scenario is outlined below.

C. CASE 1: ANALYSIS OF THE IMPACT OF AC SIDE INDUCTANCE AND DETERMINATION OF MINIMUM VALUE REQUIRED TO SATISFACTORILY LIMIT HARMONICS

The power electronics devices introduce harmonics in the system due to switching operations. According to IEEE 519 standard, %THD should not exceed the threshold of 5%. Therefore, the minimum value of AC side inductance has been evaluated by running various simulations based on various values of L. Another factor that was considered is the ability to transfer of power without being affected when increasing the AC side inductance. The minimum value of the

inductor determined through simulations has been validated analytically as explained in section IV-A.

TABLE 2. Parameters used in the model.

Parameter	Value
X/R	1.121/.044
Three phase short circuit Level (substation)	970.71 MVA
Battery Resistance	0.225Ω
Battery Capacitor	372 μF
Grid Voltage	33 kV
Inverter Voltage	480 V
Nominal voltage of BESS	960 V
Battery Capacity (Ah) /battery	149.5 Ah

1) CASE 2: ANALYSIS OF THE IMPACT OF DC SIDE INDUCTANCE DETERMINATION OF MINIMUM VALUE REQUIRED TO SUPPRESS THE IMPACT ON BATTERY CURRENT

To analyze the impact of VARs operation on the lifetime of BESS, the current to/from the battery must be monitored closely during the intervals when BESS is providing VAR support in the absence of active power. Current flow during this interval indicates that the charging/discharging cycles of the battery will be increased due to VARs operation. In order to limit this current and minimize the adverse impacts, an L-filter has been proposed at the DC side of the battery. For this purpose, the value of inductor has been determined through trial-and-error simulations with different values of inductors. This value has also been validated analytically (as explained in Section -IV-A).

2) CASE 3: WITH ZERO REAL POWER OUTPUT ($P=0$), VARY Q FROM 10% TO 100% AS A PERCENTAGE OF RATED POWER.

In this case, only reactive power is provided by the battery in the absence of real power, and monitoring of DC current from/to the battery was carried out. To achieve this objective, real power was kept at zero and the variation in the value of reactive power was done as a percent of rated power of the system, beginning from 10% to 100%. 7.5MVA is the rated power of the system.

3) CASE 4: PARALLEL OPERATION OF BESS WITH SHUNT REACTOR AND CAPACITOR BANKS ON THE AC SIDE (AT SUBSTATION)

The effect of VARs operation using BESS has been investigated by considering the already connected shunt reactors and capacitors installed at the substation for voltage support. The simulations have been carried out and the impact has been analyzed.

D. MODELING OF BESS

Fig. 3 shows the BESS system schematic connected to the grid. The battery pack is the main component of the system. A bi-directional smart inverter/ Power conversion unit

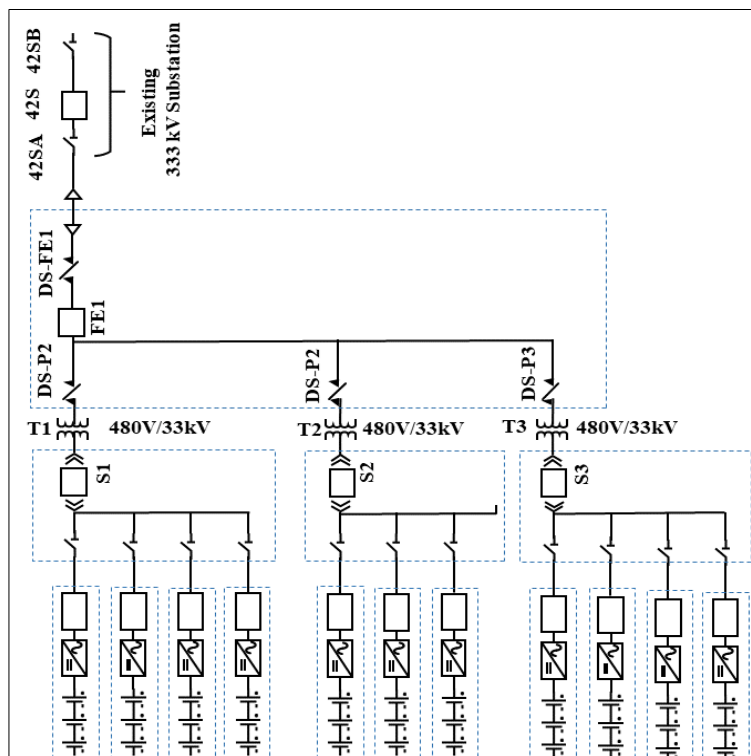


FIGURE 2. One-line of the 7.5MW/30MWh BESS.

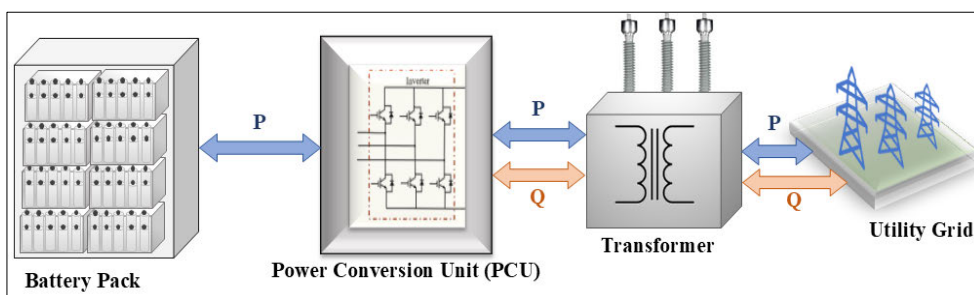


FIGURE 3. Model of BESS connected to the substation.

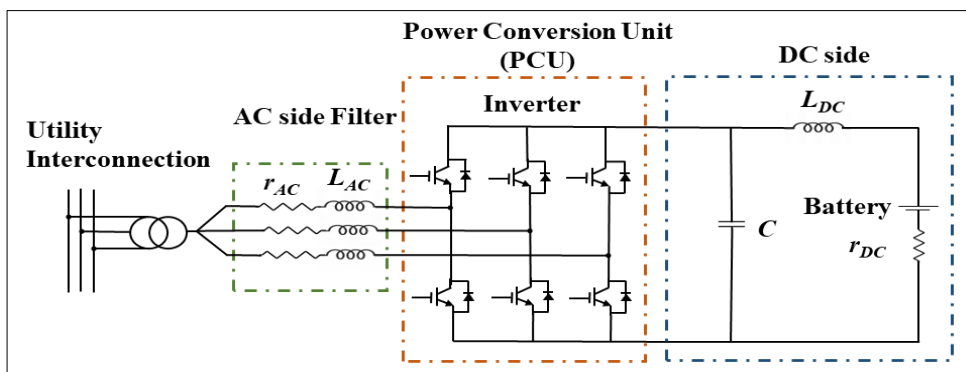


FIGURE 4. BESS configuration and inverter components.

(PCU) is used to convert AC power from the grid to DC power on the battery side, and vice versa. Since the value

of the AC output voltage of the PCU is considered 480 V, there is a requirement for a step-up transformer to step up

the inverted AC voltage at the point of common coupling. At any time instant, operation of battery can be one of these three modes which are: idle, discharging or charging. Strictly speaking, simultaneous charging, and discharging modes cannot take place. This is the case for both reactive and real powers. An approach based on vector control in the direct axis (d)-quadratic axis (q) frame (or synchronous reference frame) is used for controlling the bi-directional converter. A filter is required at the AC terminals of the converter (an L filter is depicted in Fig. 4 with its internal resistance (r_A) [38]. A Phase-Locked Loop (PLL) was used to track the angular frequency (ω). The d-q components of grid voltages and currents were obtained using Park's transformation, as further derived below. This Direct-quadrature-zero transformation (DQZ) approach is often used in order to simplify calculations for the control of three-phase inverters. The reference values of the direct-axis current (I_d) and quadratic-axis current (I_q) were computed based on desired P & Q setpoints and values of direct-axis grid voltage (V_d) and quadratic-axis grid voltage (V_q). The reference components I_d^* and I_q^* as well as measured values of I_d and I_q were fed into PI controllers to minimize the error.

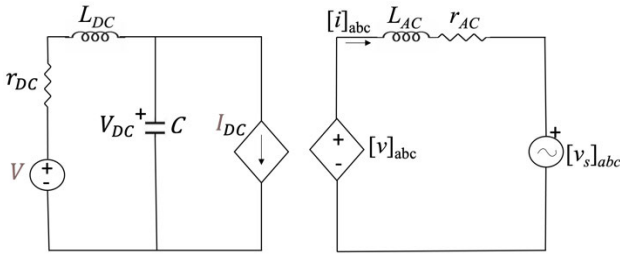


FIGURE 5. Equivalent circuit for the V_s .

To achieve independent control of real and reactive power, the dq components were decoupled. Equivalent circuits of AC and DC sides of the PCU, operated as a Voltage Source Inverter (VSI), also known as 'grid following', is shown in Fig. 5 [38]. From Fig. 5, the following equations can be derived [38],

$$C \frac{dV_{DC}}{dt} + I_{DC} = \frac{V_{DC} - V}{r_{DC}} \quad (1)$$

$$L_{AC} \frac{d[i]_{abc}}{dt} + r_{AC} i = \Delta [v]_{abc} \quad (2)$$

$$i_a + i_b + i_c = 0 \quad (3)$$

$$\Delta [v]_{abc} = [v]_{abc} - [v_s]_{abc} \quad (4)$$

where:

$[v]_{abc}$ is the inverter output voltage

$[v_s]_{abc}$ is three-phase grid voltage

Conversion from abc to dq frame of references results in the following,

$$I_d = \frac{2}{3} [i_a \cos \omega t + i_b \cos(\omega t - 120^\circ) + i_c \cos(\omega t + 120^\circ)] \quad (5)$$

$$I_q = -\frac{2}{3} [i_a \sin(\omega t) + i_b \sin(\omega t - 120^\circ) + i_c \sin(\omega t + 120^\circ)] \quad (6)$$

Similarly,

$$\Delta V_d = \frac{2}{3} [\Delta v_a \cos(\omega t) + \Delta v_b \cos(\omega t - 120^\circ) + \Delta v_c \cos(\omega t + 120^\circ)] \quad (7)$$

$$\Delta V_q = -\frac{2}{3} [\Delta v_a \sin \omega t + \Delta v_b \sin(\omega t - 120^\circ) + \Delta v_c \sin(\omega t + 120^\circ)] \quad (8)$$

Taking the derivative of (5),

$$\begin{aligned} \frac{d}{dt} I_d = \frac{2}{3} & \left[\frac{d}{dt} i_a \cos \omega t + \frac{d}{dt} i_b \cos(\omega t - 120^\circ) \right. \\ & \left. + \frac{d}{dt} i_c \cos(\omega t + 120^\circ) \right] - \frac{2}{3} \omega [i_a \sin(\omega t) \\ & + i_b \sin(\omega t - 120^\circ) + i_c \sin(\omega t + 120^\circ)] \end{aligned} \quad (9)$$

Taking the derivative of (2),

$$\begin{aligned} \frac{d}{dt} i_a &= \frac{1}{L_{AC}} \Delta v_a - \frac{r_{AC}}{L_{AC}} i_a \\ \frac{d}{dt} i_b &= \frac{1}{L_{AC}} \Delta v_b - \frac{r_{AC}}{L_{AC}} i_b \\ \frac{d}{dt} i_c &= \frac{1}{L_{AC}} \Delta v_c - \frac{r_{AC}}{L_{AC}} i_c \end{aligned} \quad (10)$$

Using (7), (8), and (10), the derivatives of (5) and (6) can be written as,

$$\frac{d}{dt} I_d = \omega L_{AC} I_q - \frac{r_{AC}}{L_{AC}} I_d + \frac{1}{L_{AC}} \Delta V_d \quad (11)$$

$$\frac{d}{dt} I_q = -\omega L_{AC} I_d - \frac{r_{AC}}{L_{AC}} I_q + \frac{1}{L_{AC}} \Delta V_q \quad (12)$$

Taking Laplace transform,

$$(sL_{AC} + r_{AC}) I_d = \Delta V_d + \omega L_{AC} I_q \quad (13)$$

$$(sL_{AC} + r_{AC}) I_q = \Delta V_q - \omega L_{AC} I_d \quad (14)$$

(13) and (14) can be arranged as,

$$\begin{bmatrix} \Delta V_d \\ \Delta V_q \end{bmatrix} = \begin{bmatrix} L_{AC}s + r_{AC} & -\omega L_{AC} \\ \omega L_{AC} & L_{AC}s + r_{AC} \end{bmatrix} \begin{bmatrix} I_d \\ I_q \end{bmatrix} \quad (15)$$

Using (4), where V_q & V_d represent the q-axis and d-axis components of the inverter output voltage, respectively, and V_{sq} and V_{sd} represent the q-axis and d-axis components of the grid side voltage, respectively.

$$\begin{bmatrix} V_d \\ V_q \end{bmatrix} = \begin{bmatrix} L_{AC}s + r_{AC} & -\omega L_{AC} \\ \omega L_{AC} & L_{AC}s + r_{AC} \end{bmatrix} \begin{bmatrix} I_d \\ I_q \end{bmatrix} + \begin{bmatrix} V_{sd} \\ V_{sq} \end{bmatrix} \quad (16)$$

The coupling between I_d and I_q can be observed from (15).

The model for the converter can be described as follows [39];

$$V_d = m_d \frac{V_{DC}}{2} \quad (17)$$

$$V_q = m_q \frac{V_{DC}}{2} \quad (18)$$

$$m_d = \frac{2}{V_{DC}}(u_d - \omega L_{AC} I_q + V_{sd}) \quad (19)$$

$$m_q = \frac{2}{V_{DC}}(u_q + \omega L_{AC} I_d + V_{sq}) \quad (20)$$

where u_q and u_d are control inputs.

After substitution of m_d and m_q in (17) and (18) from (19) and (20), it can be concluded that,

$$sL_{AC} I_d = -r_{AC} I_d + u_d \quad (21)$$

$$sL_{AC} I_q = -r_{AC} I_q + u_q \quad (22)$$

We can write the control equations as,

$$V_d = \left(K_p + \frac{K_i}{s} \right) (I_d^* - I_d) - \omega L_{AC} I_q + V_{sd} \quad (23)$$

$$V_q = \left(K_p + \frac{K_i}{s} \right) (I_q^* - I_q) + \omega L_{AC} I_d + V_{sq} \quad (24)$$

Instantaneous values of P^* and Q^* can be computed using the following equations, [40];

$$P^* = \frac{3}{2}(V_{sd} I_d + V_{sq} I_q) \quad (25)$$

$$Q^* = \frac{3}{2}(V_{sq} I_d + V_{sd} I_q) \quad (26)$$

Hence, reference values of I_d and I_q can be calculated as,

$$I_d^* = \frac{2}{3} \frac{P^* V_{sd} - Q^* V_{sq}}{V_{sd}^2 + V_{sq}^2} \quad (27)$$

$$I_q^* = \frac{2}{3} \frac{P^* V_{sq} + Q^* V_{sd}}{V_{sd}^2 + V_{sq}^2} \quad (28)$$

A schematic diagram of the inverter controller is shown in Fig. 6.

Simulated Cases:

The proposed configuration has been utilized to simulate the following cases without and with the DC side inductor (L_{DC}). L_{DC} is connected between the converter and battery, as illustrated in Fig. 4.

Case 1: Analysis of the impact of inductance at the AC side and determination of minimum value required to limit harmonics satisfactorily.

Case 2: Analysis of the effect of DC side inductance and determination of minimum value required to suppress the impact on battery current.

Case 3: With zero real power output ($P = 0$), the value of Q is varied as a percentage of rated apparent power (10% to 100%).

Case 4: Parallel operation of BESS with shunt capacitor and reactor banks on the AC side (at substation).

The power flow in/out of the battery cannot exceed the maximum value of rated apparent power as stated by the smart inverter and the associated transformer. The apparent power of battery system is a nonlinear function of the

exchanged real and reactive power, as simply expressed in (29), as previously depicted in Fig. 1 above.

$$\sqrt{P^2 + Q^2} \leq S \quad (29)$$

For each of the real (P) and reactive (Q) powers, one variable is designated while the battery defines two variables for each discharging and charging, given its bi-directional nature (PB^c , PB^d , QB^c , QB^d). Because, at any time instant either discharging or charging real powers can have a non-zero value [27]. The same is also applicable to reactive power, the net apparent power can be expressed as the summation of the discharging and charging powers using (29). Subsequently, the flow limit can be written as (30).

$$\sqrt{(PB_{\beta,k}^c + PB_{\beta,k}^d)^2 + (QB_{\beta,k}^c + QB_{\beta,k}^d)^2} \leq SB_{\beta}^{rated} \quad \forall \beta \in \beta, \quad \forall k \in K \quad (30)$$

where

- K: set of time periods
- β : set of batteries within the system
- SB_{β}^{rated} : Apparent Power rating (VA) of battery β
- $QB_{\beta,k}^D$: reactive discharging power (VAR) of battery β at time k
- $QB_{\beta,k}^C$: reactive charging power (VAR) of battery β at time k
- $PB_{\beta,k}^D$: active discharging power (W) of battery β at time k
- $PB_{\beta,k}^C$: active charging power of battery β at time k (W)

Note that the AC filter, as shown in Fig. 4, is intended to reduce the high-frequency harmonics resulting from pulse width modulation, as commonly embedded in smart inverters.

III. RESULTS AND DISCUSSION

A. CASES 1 AND 2. IMPACT OF DC AND AC SIDE INDUCTANCES

The charging, injection, or flow of power either real or reactive power into the battery is denoted by a positive sign. Negative sign indicates the discharge or flow of powers (either real or reactive powers) out of the battery.

Similarly, the negative and positive signs of I_{DC} imply the charging and discharging of BESS, respectively. The effect of reactive power support on the lifetime of the battery system can be examined by monitoring the DC current (I_{DC}), particularly, during the time intervals when the battery system is supplying reactive power (real power is zero i.e., $Q \neq 0$ and $P = 0$). In this paper, and as shown in the plotted results, the intervals of time from 3-4s and 7-8s pertain to the scenarios where there is no flow of active power to or from the BESS whereas reactive power flows to and from the battery respectively. I_{DC} has been observed in these three modes of operation of inverter: [$P \neq 0$, $Q \neq 0$], [$P \neq 0$, $Q = 0$] and [$P = 0$, $Q \neq 0$].

Inductance at the DC side not only affects the thermal losses but also the quality and magnitude (oscillations and

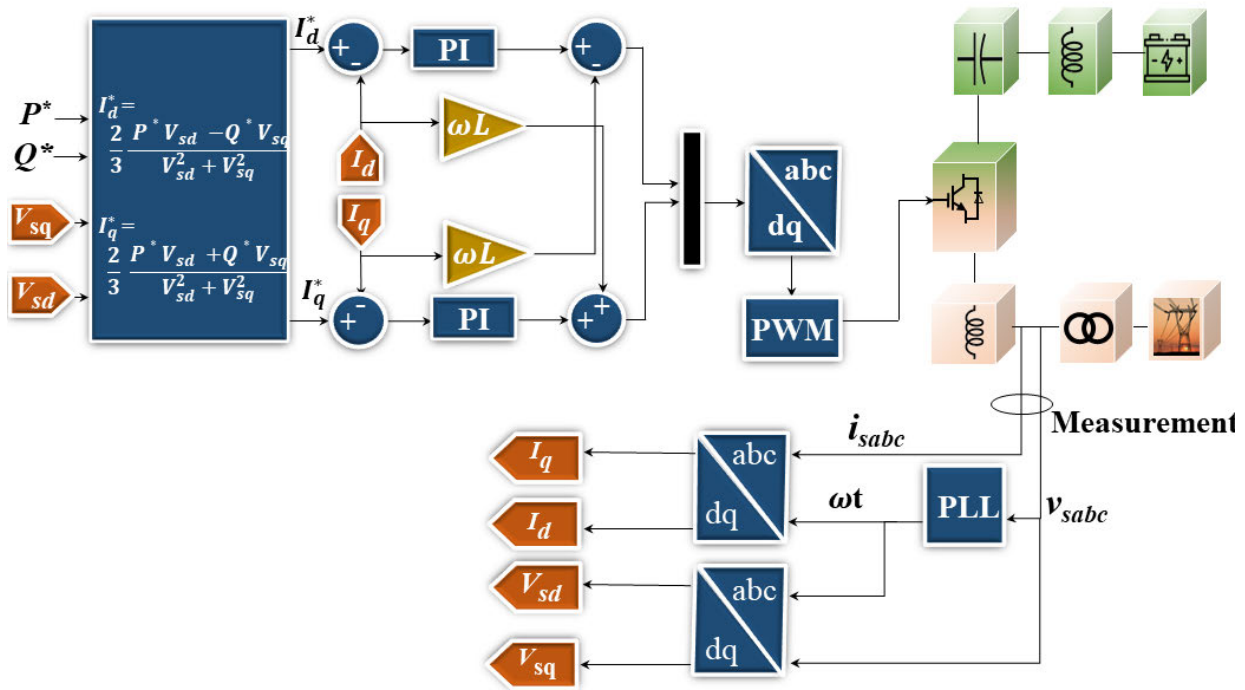


FIGURE 6. Schematic diagram of the inverter controller.

ripple) of battery current. To minimize the effect of VAR support on lifetime of battery, the flow of current through BESS should be zero in the ideal case during the mode of operation where $[P = 0, Q \neq 0]$. Subsequently, there is a need to determine the optimal value of DC side inductance which can restrain the flow of direct current without having adverse effects on other operations or performance of BESS.

Further, the AC side inductor filters out the ripples/oscillations in the AC current and maintains the value of Total Harmonic Distortion (THD) within allowable limits, (i.e., $\text{THD} < 5\%$ as per IEEE standard 519). However, due diligence is required while selecting the inductance value. A high filter value on the AC side can delimit the power transfer capability and increase system losses. A more optimal design of filter depends on the minimum value of AC side inductance which limits the THD value below 5%. For this purpose, we simulated various cases while keeping L_{DC} constant ($L_{DC} = 0.25\text{mH}$), whereas, the value of L_{AC} was increased gradually (initial value of $L_{AC} = 1\text{mH}$) and the value of THD was monitored. For the simulation of this scenario i.e., determining the minimum value of L_{DC} and L_{AC} , the apparent power and power factor of BESS is considered as 7.5 MVA and 0.9 respectively.

The THD was observed to be less than 5% for the value of $L_{AC} = 5\text{mH}$, Figs 7 and 8 illustrate the simulation results for L_{AC} values of 1mH and 5mH, respectively. Both figures have been divided into seven subplots. The first subplot depicts the power to/from the battery while the second subplot shows the apparent power. The third plot shows the AC side voltage

and current on the AC side. An amplified view of voltage and current on the AC side is shown below for better visualization. The analysis of the 1st and 3rd subplots along with the angle difference between voltage (V_a) and current (I_a) on the AC side is tabulated below;

It can be observed from Table 3 that I_a is lagging V_a on AC side when reactive power is absorbed by the battery system i.e., $Q > 0$. On the other hand, when $Q < 0$ i.e., reactive power is being discharged by the battery system, I_a is leading V_a . However, V_a and I_a are in phase and out of phase with each other when reactive power is zero while P is absorbed and discharged by the battery, respectively.

TABLE 3. Subplots analysis of Figs 7 & 8.

Interval (s)	P & Q direction	Angle Difference between V_a & I_a
0-1	$P=0, Q=0$	$I_a=0$
1-2	$P>0, Q=0$	V_a & I_a are in phase
2-3	$P>0, Q>0$	I_a lags V_a
3-4	$P=0, Q>0$	I_a lags V_a
4-5	$P<0, Q>0$	I_a lags V_a
5-6	$P<0, Q=0$	180 out of phase (P is flowing from battery to grid side)
6-7	$P<0, Q<0$	I_a leads V_a
7-8	$P=0, Q<0$	I_a leads V_a
8-9	$P>0, Q<0$	I_a leads V_a
9-10	$P=0, Q=0$	$I=0$

The fifth subplot pertains to the percentage of THD. THD value depends upon the amount of P . Battery voltage is high

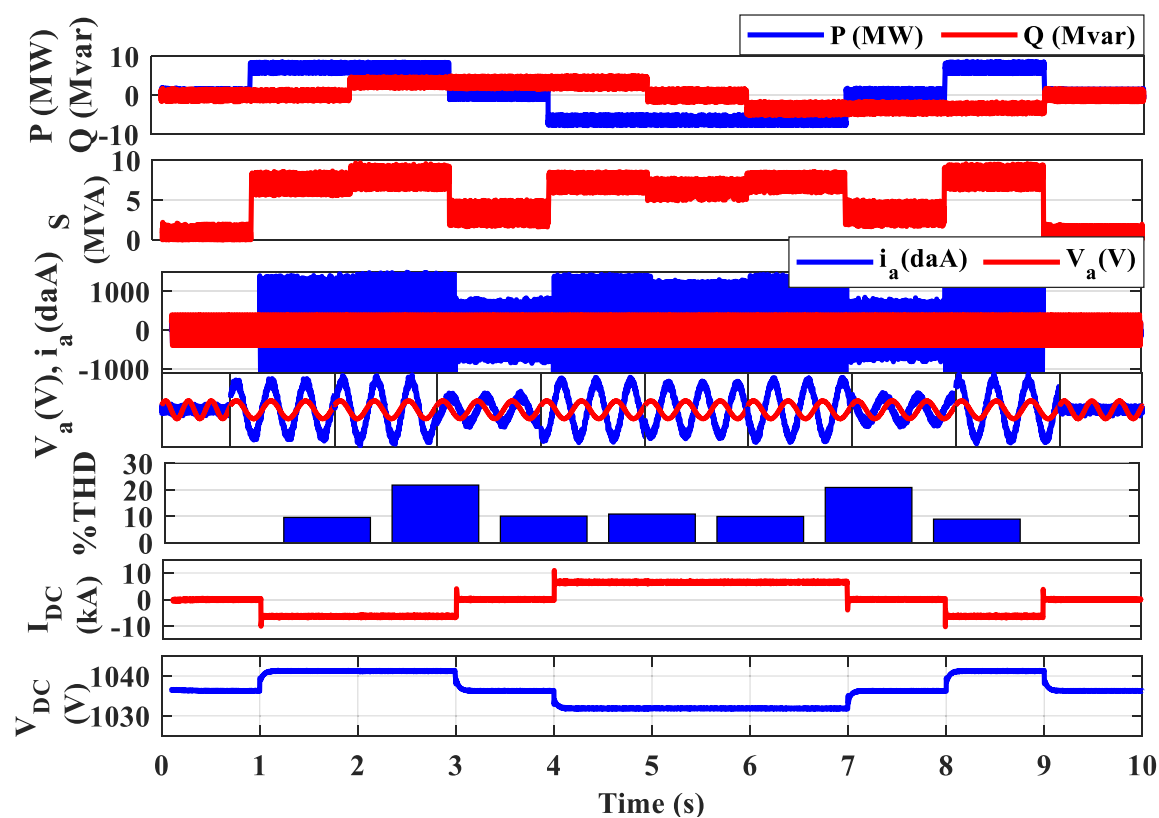


FIGURE 7. Results for $L_{DC} = 0.25 \text{ mH}$ & $L_{AC} = 1 \text{ mH}$.

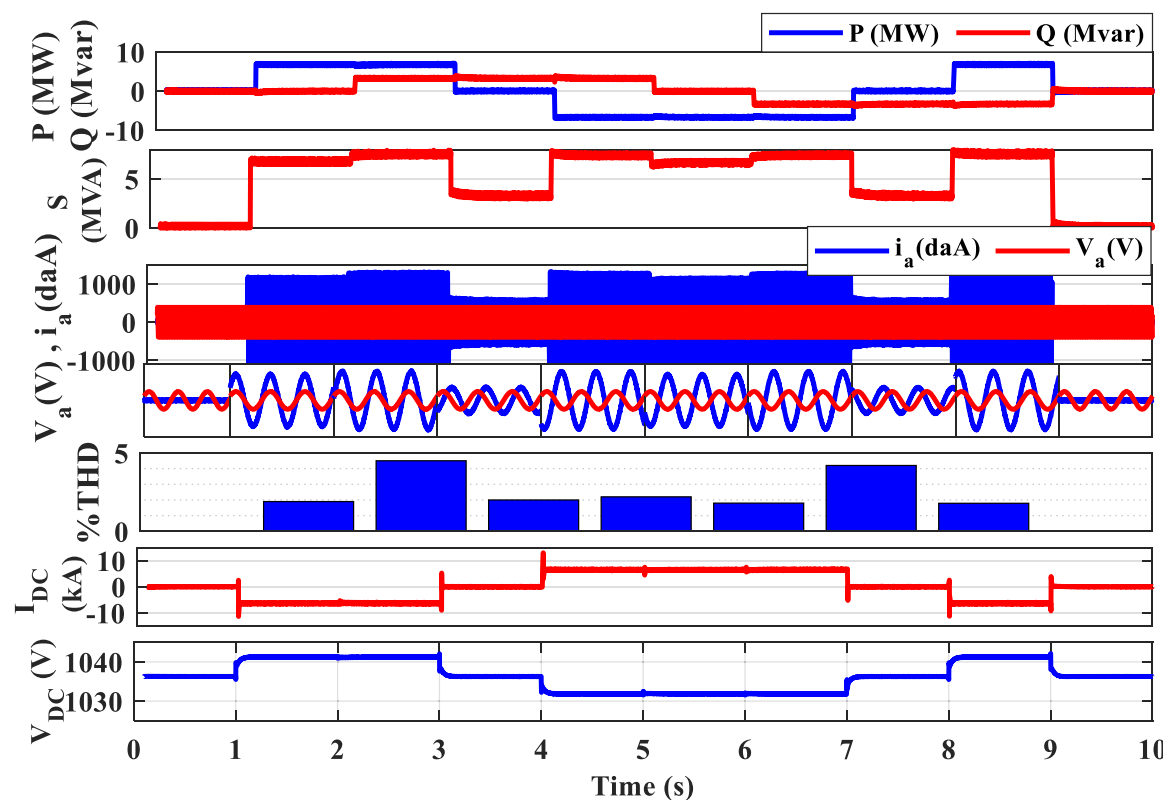
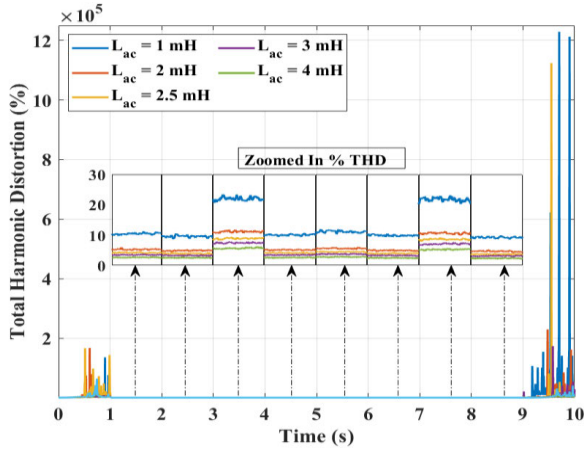
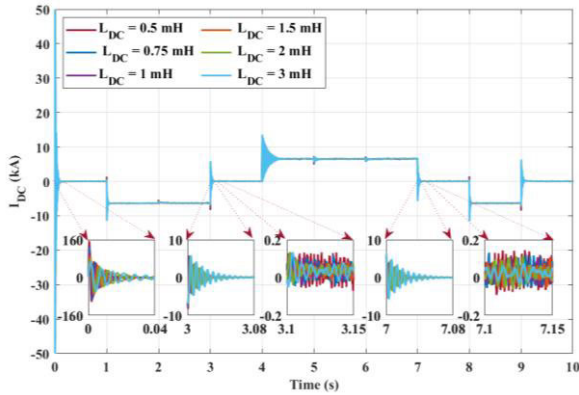


FIGURE 8. Results for $L_{DC} = 0.25 \text{ mH}$ & $L_{AC} = 5 \text{ mH}$.

FIGURE 9. %THD vs L_{AC} .FIGURE 10. I_{DC} for various values of L_{DC} .

(1041V) during the intervals when P is absorbed (1-3s and 8-9s) and goes low till the 1031V when $P < 0$ i.e., the battery is discharging (4-7s). When $P = 0$, voltage remains 1038V (during 0-1s, 3-4s, 7-8s, 9-10s).

Fig. 9 summarizes the variation in values of THD with various sizes of L_{AC} . The highest values of THD are found to be 4.8%, 5.3%, 5.8%, 7.4%, 11.4%, and 22.6 % for corresponding values of L_{AC} 5, 4.5, 4, 3, 2, 1mH respectively. It is worth mentioning that both reactive and real powers are zero during the last and first intervals, subsequently, it is meaningless to compute or analyze the THD values during these intervals. Spikes in THD are naturally observed for short instants of time during inter-interval transitions. These correspond to the high-frequency content associated with the transitioning signal from one value to another. This is understandably a computation shortage since the block computes continuously and does not reflect a real detrimental impact on the system.

The desired margin for the current ripple can be regulated by calculating the value of DC-side inductance. This is achieved using a specific formula that equates the increase in energy of the inductor during its conduction phase with the

decrease in energy during the inductor's discharge phase.

$$L_{DC} = \frac{V_{IN}(V_{OUT} - V_{IN})}{\Delta I_L f_s V_{OUT}} \quad (31)$$

$$\Delta I_L (\text{Current Ripple}) = 0.2 I_{OUT(max)} \frac{V_{OUT}}{V_{IN}} \quad (32)$$

With f_s (switching frequency) = 7980 Hz, $V_{OUT} = 1037$ V, $V_{IN} = 960$ V, and $I_{OUT(max)} = 41$ A (measured), the current ripple $\Delta I_L = 8.86$ A and as a result, $L_{DC} = 0.001008$ H which is the minimum value. Hence, the value found in Section II using the trial and error method, came out to be 1mH.

The energy exchange with grid in half-cycles is ensured by the AC side inductance and the impact on DC side is buffered. Computation of AC side inductance (L_{AC}) can be computed by using (33) [42].

$$L_{AC} = 3V_s I_{c1} = 3V_s \frac{V_{c1}}{\omega Q_{c1}} \left(1 - \frac{V_s}{V_{c1}}\right) \quad (33)$$

Using V_{c1} (fundamental component of voltage at inverter side(measured)) = 629.6 V, V_s (voltage at grid-side) = 480 V, Q_{c1} = maximum fundamental component of Q = 299, I_{c1} = fundamental component of current at inverter side, $f = 60$ Hz (ω = angular frequency), the value of inductance at AC side (L_{AC}) comes out to be 9 mH.

The value of L_{AC} found via simulations can be verified by comparing it with the value calculated above. However, the computed value is higher because the inductance model took into consideration the inductor's internal resistance, the reduction in current also depends on it. After establishing the minimum size of the AC side inductor (5mH), the value of L_{DC} gradually increased. It was noted that increase in L_{DC} reduced the oscillations during 3-4s and 7-8s. However, a further increase in the value of L_{DC} beyond 1mH, gives rise to the high ripples/oscillation in other intervals as shown in Fig. 10. As a result, we selected 1mH as an appropriate value of L_{DC} . The results with $L_{DC} = 5$ mH and $L_{AC} = 1$ mH are shown in Fig. 11.

B. CASE 3: STEP RESPONSE IN FOUR QUADRANTS

Initially, the battery does not supply any real power, focusing solely on providing reactive power, while the DC current to and from the battery is monitored. To achieve this, P was set to zero, and the value of Q was adjusted as a percentage of the system's rated power, ranging from 10% to 100%. The system's rated power is 7.5MVA. From this setup, it can be observed from Fig. 12 (Subplot number. 1) that the value of current to or from the battery remains approximately 1% of its rated current when VARs are being injected and absorbed, as observed in the simulation results where P is maintained at 0 throughout all intervals. Initially, Q is at its maximum, or 100% (7.5 MVAR), from 0s to 1s. Subsequently, Q 's value decreases to 75% (5.625, MVAR), 50% (3.75 MVAR), 25% (1.875 MVAR) and 10% (0.75 MVAR) for the time intervals of 1-2s, 2-3s, 3-4s, and 4-5s, respectively, indicating the battery is absorbing Q during these first 5 seconds

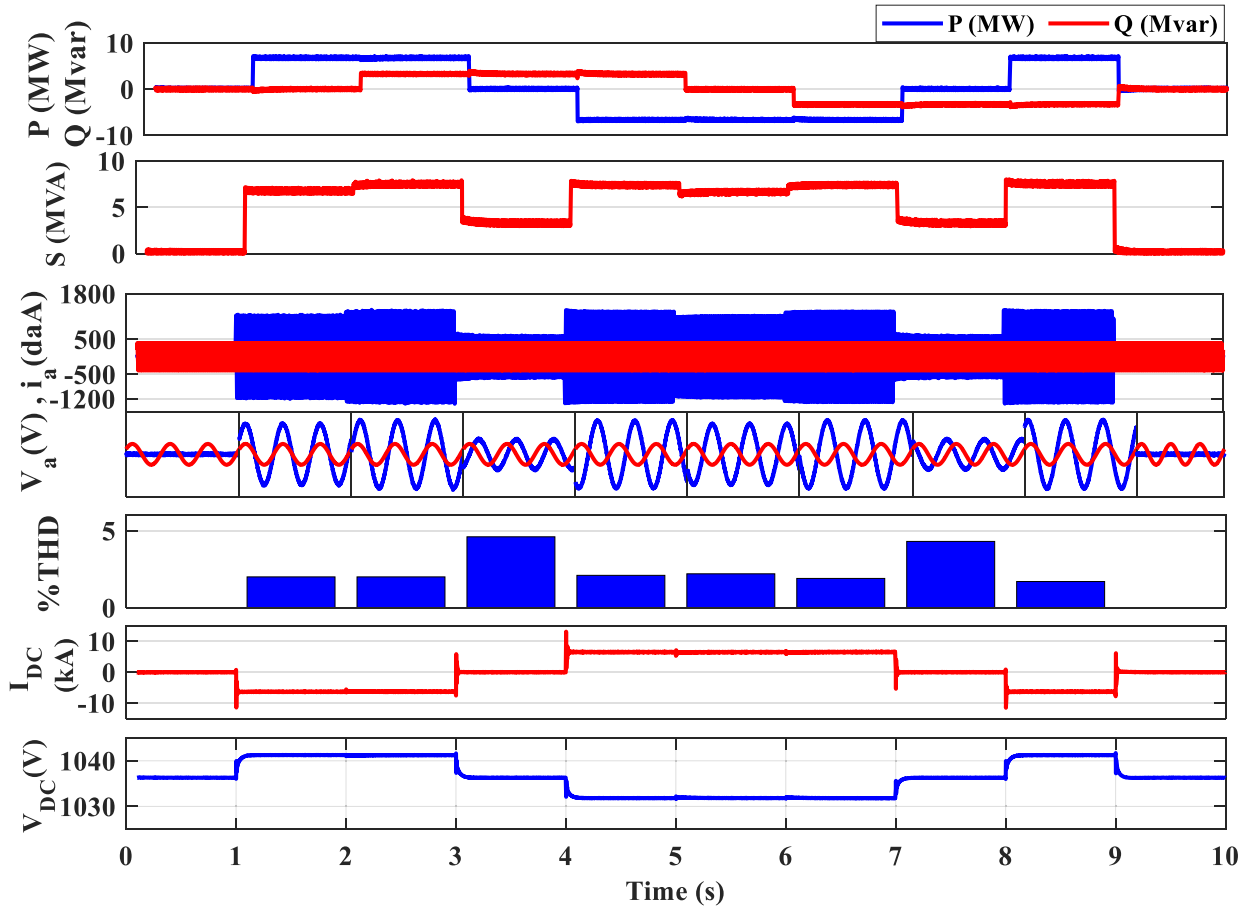


FIGURE 11. Outcomes of $L_{DC} = 1\text{mH}$ & $L_{AC} = 5\text{mH}$.

of the simulation. Following this, Q drops to -10% in the interval of 5-6s, and for the intervals of 6-7s, 7-8s, 8-9s, and 9-10s, Q values are adjusted to -25% , -50% , -75% , and -100% respectively. It is clear from Fig. 12 that the Total Harmonic Distortion (THD) surpasses 5% during the intervals of 3-4s, 4-5s, 5-6s, and 6-7s, when Q is at 25% , 10% , -10% , and -25% , respectively. Section III in Fig. 12 highlights the AC side's voltage and current; during the 0-5s interval, Q is positive, indicating the battery absorbs reactive power, and the current on the AC side lags behind the voltage. Conversely, in the next 5 intervals from 6-10s, Q is negative as the battery discharges reactive. The battery system discharges reactive power, resulting in the current leading to the voltage on the AC side. According to subplot No. 6, the current flowing to or from the battery is consistently negligible, and the battery's output voltage remains stable at approximately 1037V , as seen in subplot 7. Furthermore, subplot No. 5 in Fig. 12 indicates that the Total Harmonic Distortion (THD) stays below a 5% threshold when Q is at $+50\%$. It's important to highlight that in scenarios 1 and 2, where the AC side inductance value was calculated, the assumption was made that the battery system operates at a power factor of 0.9 . Thus,

the THD remains at 5% provided it does not surpass the Q value that aligns with a 0.9 power factor.

Fig. 13 presents the percentage of THD relative to different L_{AC} values for either absorbing or injecting reactive power (VARs) at $Q = 25\%$. The required inductance value on the AC side is identified as 8.5mH , as illustrated in Fig. 13. The analysis further extends to determining the L_{AC} value for $Q = 10\%$, as shown in Fig. 14 the value of $L_{AC} = 21\text{mH}$, the THD would be less than 5% if the battery needs to inject or absorb reactive power (VARs) in absence of active power, i.e., $P = 0$.

The impact of VAR support on the health of the battery has been calculated by using the following Index (I_{SoH}) [42].

$$I_{SoH}(\%) = \frac{\int_{t_1}^{t_2} I_{battery}}{I_{rated}(t_2 - t_1)} \cdot 100 \quad (34)$$

This is the ratio of maximum charge that can be released over the I_{rated} (rated battery current). $I_{battery}$ is the current flowing through the battery during VAR support (in absence of active power). The value of I_{SoH} comes out to be 4.66% indicating that VAR support costs about 4.66% of the battery

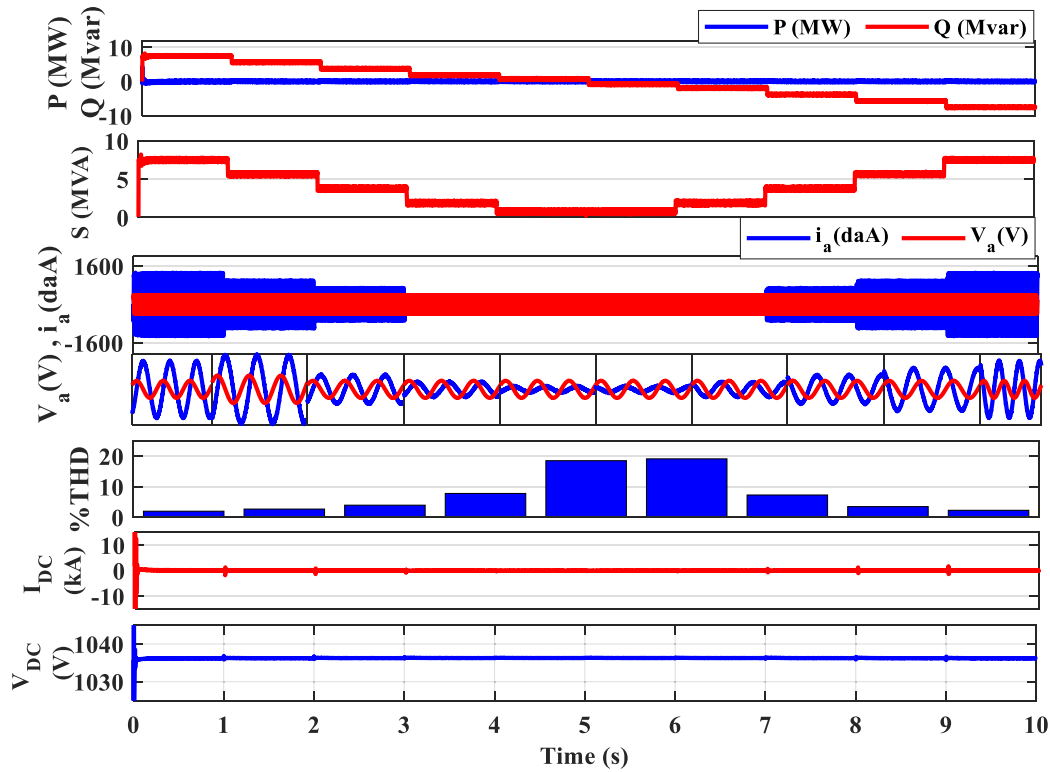


FIGURE 12. Results for $P = 0$ Vs Different values of Q .

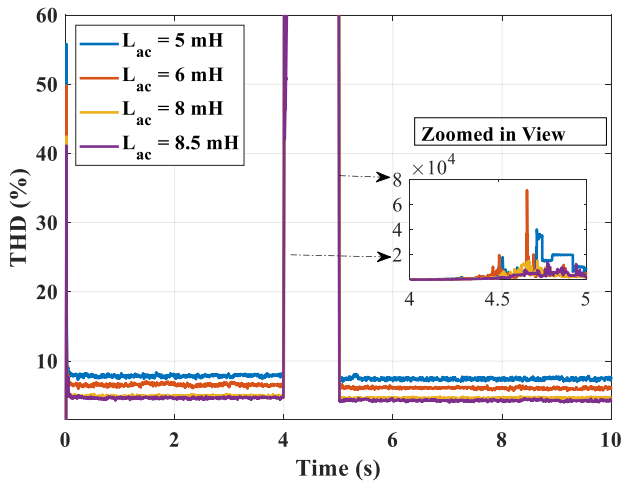


FIGURE 13. % THD for different values of L_{AC} ($Q = 25\%$).

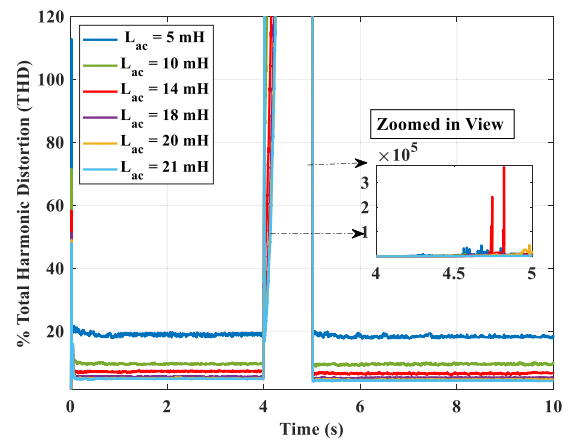


FIGURE 14. THD Vs several values of L_{AC} ($Q = 10\%$).

lifetime. The results from cases 1-3 indicates that with the proper inductive filters in place, VARs compensation have no implications on the degradation of battery.

C. CASE 4: IMPACT OF CAPACITOR BANKS AND SHUNT REACTORS ON THE AC SIDE

This scenario is divided into two segments where we simulate and assess the effect of supplying VARs from the battery

system alongside capacitor banks/shunt reactors, mirroring real-world grid functionalities at the substation. We have modeled the specific capacitor banks and shunt reactors employed by a utility company in New York City. The capacitor banks, with a capacity of 10 MVAR, and shunt reactors, with a capacity of 1 MVAR, were individually connected to the AC side through a Y-connection with a neutral point. The findings of this analysis are presented in Figs 15 to 20.

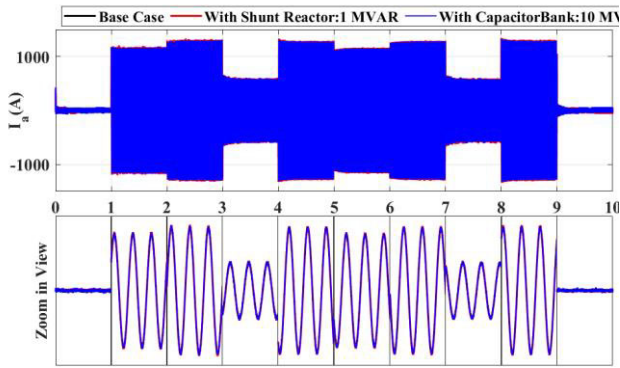


FIGURE 15. I_{AC} used for 3 scenarios.

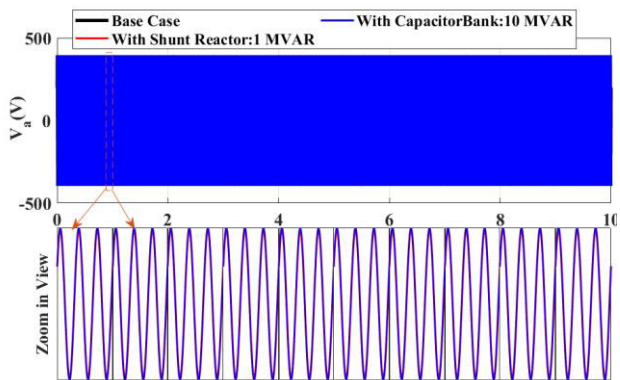


FIGURE 16. V_{AC} for 3 scenarios.

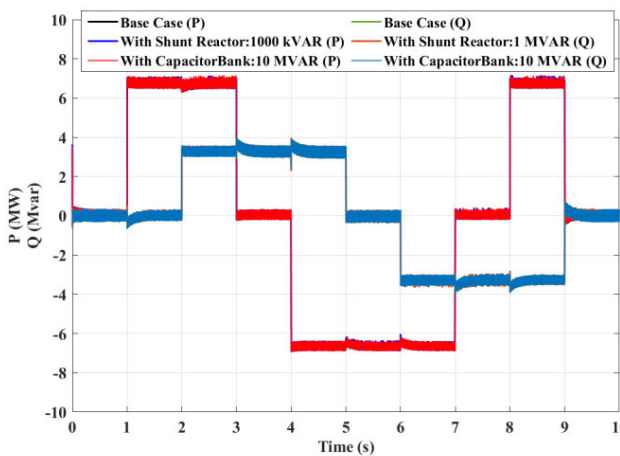


FIGURE 17. P & Q for 3 scenarios.

The effects of adding shunt reactors and capacitor banks are evident in the AC side current, voltage, and both active and reactive power, as illustrated in Figs. 15, 16 and 17. The percentage of Total Harmonic Distortion (%THD) is shown in Fig. 18. Figs 19 and 20 display the current and voltage on the DC side, respectively. When comparing the baseline scenario

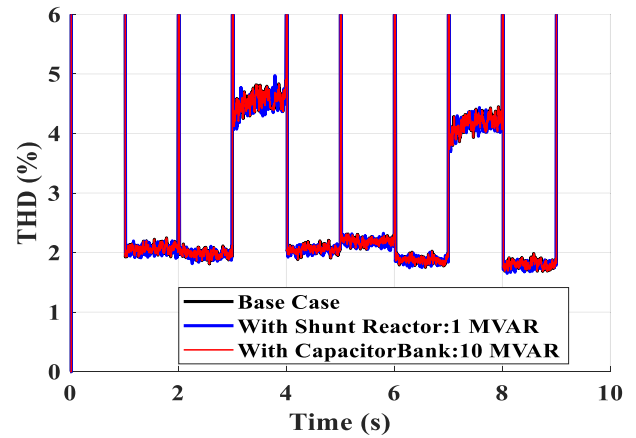


FIGURE 18. %THD for 3 scenarios.

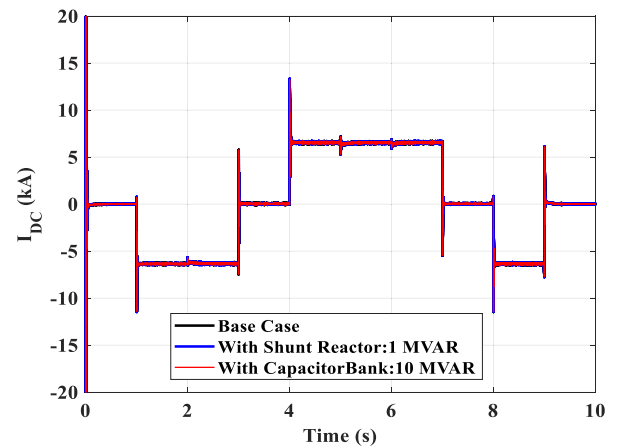


FIGURE 19. I_{DC} used for three scenarios.

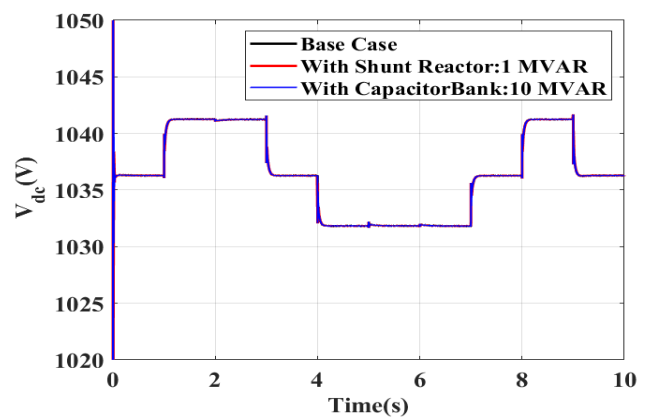


FIGURE 20. VDC for three cases.

(the case without any shunt reactor or capacitor connected to the AC side) with instances involving shunt reactors and capacitor banks, it appears that the addition of capacitor banks or shunt reactors has neither a positive nor a negative impact on the BESS's ability to provide reactive power.

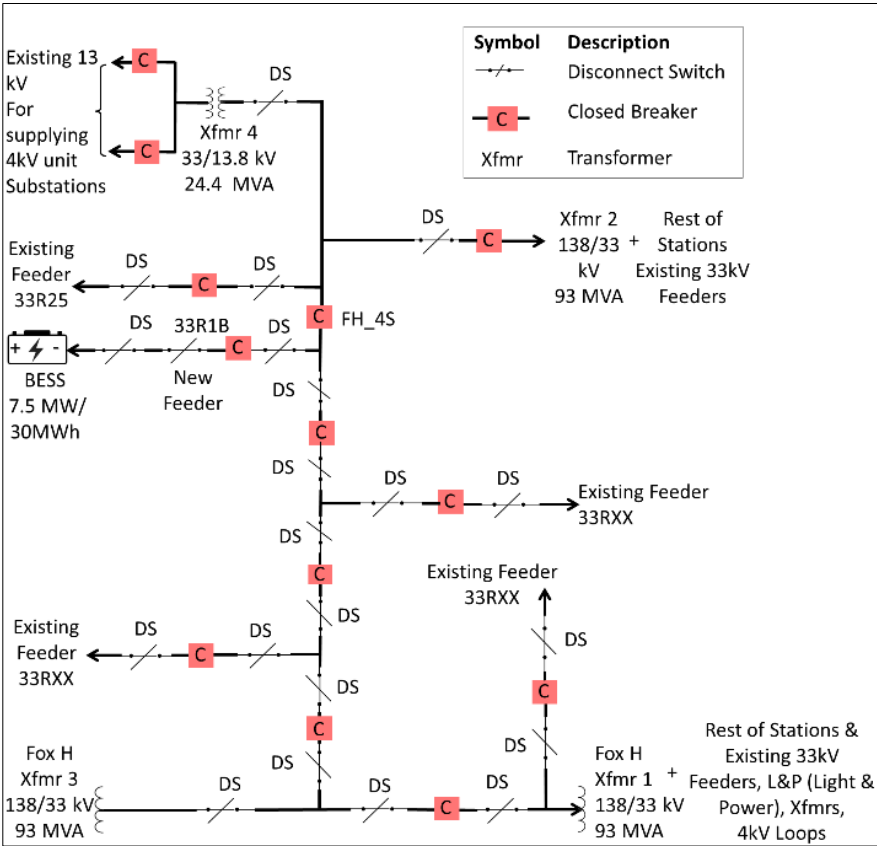


FIGURE 21. Interconnection diagram of BESS.

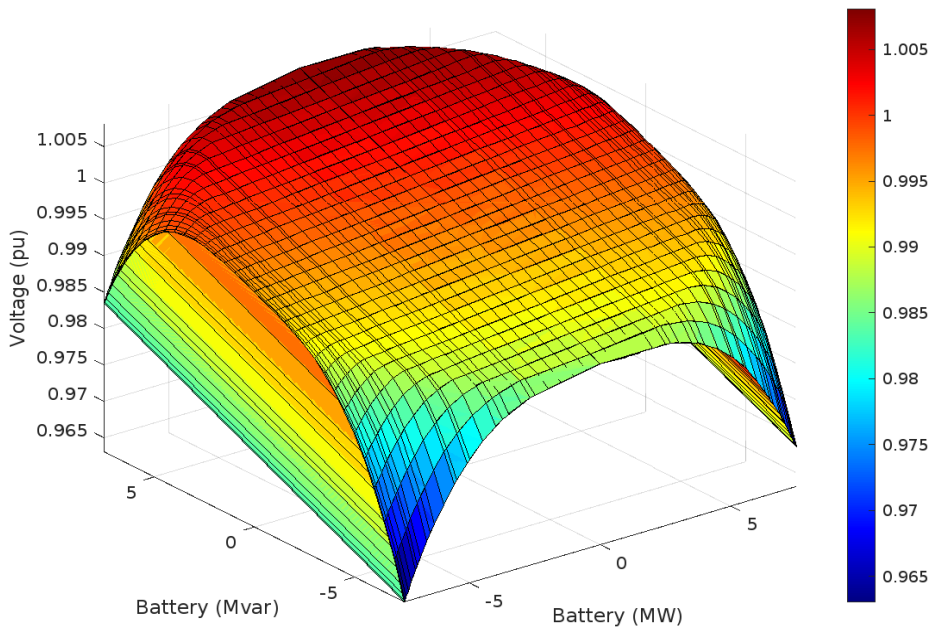


FIGURE 22. Effect of P & Q flows from battery on voltage.

While traditional grid devices such as capacitor banks, shunt reactors, and STATCOM are effective at providing voltage support, the drawback is that these are binary systems,

that is, either on or off at the full rated capacity. On the other hand, BESS allows operators to tailor the VARs response to evolving grid conditions without installing or maintaining

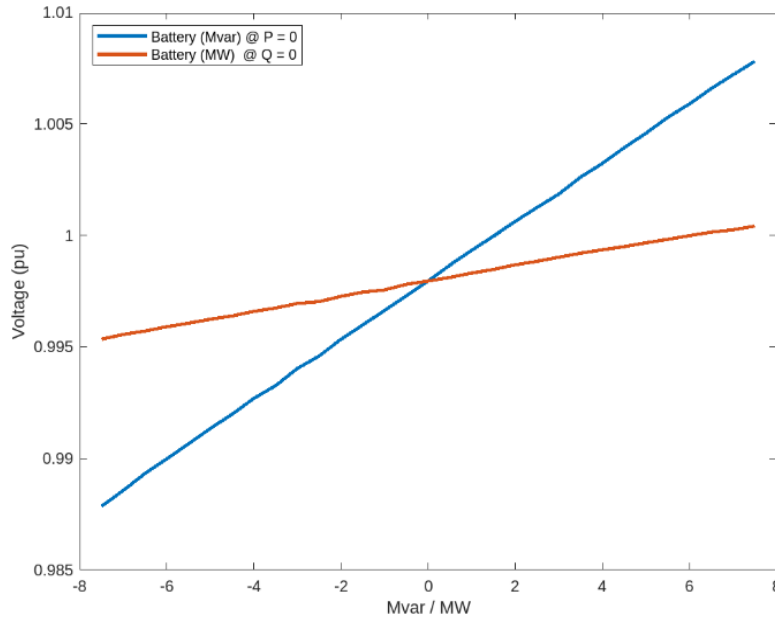


FIGURE 23. Battery individual effects of P&Q on voltage.

additional equipment. Hence, with a single BESS installation, installation of STATCOM can be avoided and the same functions can be achieved year-round vs. certain times of the year. Furthermore, the BESS will not have to be resized or the warranty structure will not have to be renegotiated beyond the active power requirements.

IV. LOAD FLOW ANALYSIS

A. DESCRIPTION OF THE CASE

The 7.5MW/30MWh BESS under consideration will be interconnected directly to a Con Edison 34.5kV area substation. This area substation steps down 138kV transmission voltage to 33kV primary distribution voltage and then supplies this lower voltage to various distribution systems components. Within the network/load area, they are 14– 33kV and 2– 13kV primary supply feeders and 77 network transformers ensuring continuity of power supplied to the distribution system serving approximately 200 MW load. Moreover, the station is equipped with six shunt reactors, with capacities ranging from 600 to 1000 kVAR. These reactors are essential for providing reactive power compensation and ensuring voltage stability at the station.

As outlined in the methodology section, the BESS is set to be interconnected to the 42-S breaker position via a new 33kV feeder, labeled 33R16. This arrangement will directly integrate the system with the substation's "ring bus," facilitating power distribution across the existing primary feeders. The feeder 33R29, being nearest to breaker 42-S and electrically closest, will connect to 33R16 through a newly established FH_4S bus. This bus will then link to the rest of the ring bus through additional 33kV breakers. This setup will serve as the basis for modeling various network load conditions (at the network's independent peak) to evaluate the BESS's

effectiveness in ensuring network stability. Fig. 21 illustrates the BESS's one-line interconnection with the substation ring bus. This BESS interconnected with Con Edison's existing network/substation has been modelled and simulated in Open DSS. The rest of the network has also been modelled. The 24-hour load has been computed by calculating the average of the load for the whole year (i.e., average load profile of 365 days).

B. LOAD FLOW RESULTS AND DISCUSSION

Fig. 22 illustrates the effects of varying the set-points for real power (P) and reactive power (Q) on the voltage level at FH_4S. Discharging of the battery system, indicated by positive values for P (MW) and Q (MVAR), contributes active or reactive power to the network. However, negative values signify the battery system's charging phase, where it absorbs active or reactive power.

At the FH_4S bus, the voltage fluctuates within a range, with a maximum of 1.008 per unit and a minimum of 0.9651 per unit. The voltage magnitude peaks when the battery delivers its maximum reactive power (7 MVAR) and active power into the grid. The voltage was not confined to the standard upper limit of 1.05 per unit intentionally, to showcase the system's capacity for voltage compensation under scenarios of increased load beyond the simulated current demand in this study.

Furthermore, Fig. 23 confirms that VAR compensation plays a more significant role in maintaining the voltage level. As a base case, the simulations were carried out without any reactive power i.e., ($Q = 0$), and then setting active power to zero ($P = 0$). Several other scenarios were tested at varying P & Q set-points to reflect real-time conditions. As shown, the voltage curve is steeper as VARs are injected,

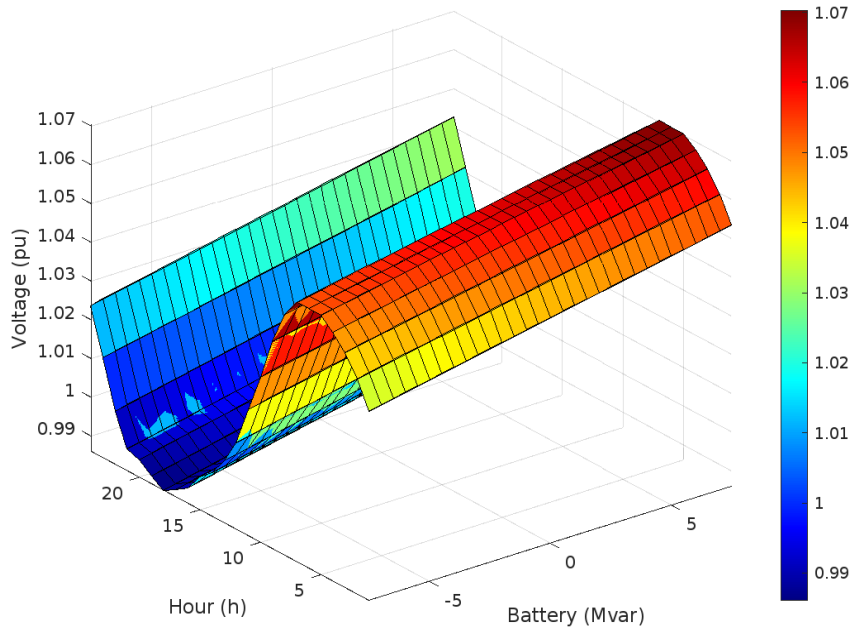


FIGURE 24. Effect of battery P&Q on 24-hour voltage profile.

which establishes that the battery can potentially influence the voltage magnitude at the substation (support reactive power). Whether it is during a voltage drop for excessive load or a voltage rise during high renewable production, the impact of VARs operation on the voltage level is higher than the impact of active power operation. Fig. 24 shows the impact of absorbing and injecting VARs during a typical 24-hour load at the modeled substation. Throughout the day the voltage level varies based on the changing demand (load). Controlling the battery Q set-points (from -7.5 MVAR to 7.5 MVAR) can aid in maintaining voltage stability through transitions from off-peak to peak loads. By employing the $-Q$ set points (which involve absorbing MVAR), it's possible to control or regulate the bus voltage, thereby assisting in managing overvoltage conditions that often occur with high solar PV penetration, particularly during the hours of 10:00 – 15:00. Similarly, during high load demand or peak hours (16:00 – 19:00), $+Q$ setpoints (VAR injection) can support under voltage conditions at the substation and help mitigate the need for operational measures such as preemptive 5-8% voltage reduction.

V. CONCLUSION

The values of filtering elements on the AC and DC sides play a crucial role in maintaining the %THD and DC current within acceptable thresholds. Thus, when designing the BESS, it is essential to carefully consider the use cases as examined. For the Con Edison battery scenario, with an assumed apparent power of 7.5MVA and a power factor of 0.9, findings indicate that the minimum AC side inductance required is 5mH to maintain THD below 5% according to IEEE standards. Similarly, the minimum inductance needed on the DC side is estimated to be 1 mH. If the threshold is

surpassed, limit cycle oscillations will intensify. Simulations have been conducted where the active power is maintained at zero, and reactive power is adjusted in increments from its maximum of 100% (7.5 MVAR) down to 10%. These simulations demonstrate that the BESS can manage reactive power fluctuations without an increase in battery current, thus maintaining the State of Charge (SOC) and minimizing degradation. Nonetheless, for additional reactive power support exceeding a power factor of 0.9 (beyond the designed operational range), our models indicate a need to reassess (and potentially increase) the AC side inductance to maintain Total Harmonic Distortion (THD) within acceptable levels. Furthermore, our findings indicate that the operation and lifespan of the battery remain unaffected when it operates alongside shunt reactors and/or capacitor banks on the AC side.

The outcomes of steady state/load flow analysis reveal that the battery's Q set points can be effectively used for voltage regulation, particularly during the grid's critical operational phases. The battery also plays a role in modulating the substation's voltage magnitude, aiding in reactive power compensation during scenarios of voltage dips due to high demand or voltage spikes from abundant renewable energy generation. Notably, the influence of the BESS's VAR operations on the substation bus voltage profile is more pronounced than that of its real power contributions.

Given the scenarios tested within this study, BESS emerges as a feasible solution for providing VAR support to the grid. Importantly, offering VAR support does not adversely affect the battery's lifespan more than energy provision does. It's recognized that battery cycling, whether for VAR support or energy provision, will inherently reduce its lifespan. The study can be extended to analyze the various other

applications of BESS and their impact on the actual distribution network.

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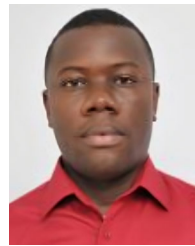
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